

Michael Nevels

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Professional Summary

Senior Software Engineer with over 10 years of experience architecting and building healthcare technology platforms that drive measurable impact across patient engagement, care coordination, claims optimization, and provider performance analytics. Deep expertise in C#/.NET 8, ASP.NET Core, Python (FastAPI, Django, Flask), Angular, React, SQL Server, Azure, and Kubernetes, with strong knowledge of FHIR/HL7 interoperability, HIPAA compliance, and event-driven microservice architecture. Proven ability to own core product areas end-to-end while collaborating closely with cross-functional stakeholders — clinicians, compliance officers, data scientists, and product managers — to ensure solutions are usable, secure, scalable, and aligned with regulatory and business goals. Known for roadmap facilitation, stakeholder workshops, and mentoring engineering teams, as well as leveraging AI-assisted development tools, MLOps pipelines, and modern DevOps practices to accelerate delivery and improve quality.

Core Skills

- Languages & Frameworks: C#, .NET 8, ASP.NET Core, Python (FastAPI, Flask, Django), Angular 17, React 18, Redux, RxJS, Entity Framework Core, LINQ, gRPC, GraphQL, REST APIs
- Cloud & DevOps: Azure App Services, Cosmos DB, Azure Service Bus, Azure Kubernetes Service (AKS), AWS ECS/Fargate, Docker, Kubernetes, Terraform, Helm, GitHub Actions, Azure DevOps, ArgoCD
- Healthcare Expertise: FHIR, HL7 v2, EHR Integration (Epic, Cerner), Care Coordination, Claims Processing, CMS/HEDIS Quality Reporting, Patient Risk Scoring
- AI/ML & Data: MLflow, Vertex AI, Hugging Face, Pandas, PySpark, Airflow, LangChain, GPT-based copilots, model monitoring, feature engineering pipelines, data lineage tracking
- Security & Compliance: HIPAA, PHI/PII Handling, SOC2, OAuth2, OpenID Connect, JWT, ABAC/RBAC, Azure Entra ID, Key Vault, TLS, AES-256, Vulnerability Scanning
- Testing & Quality: xUnit, NUnit, Pytest, Cypress, Playwright, Selenium, Postman, k6, SonarQube, Trivy, Gitleaks, Contract Testing, Mocking Frameworks
- Soft Skills: Stakeholder Communication, Roadmap Planning, Facilitation of Architecture Reviews, Technical Writing, Mentorship, Conflict Resolution, Change Management

Professional Experience

One Medical

Senior Software Engineer | 07/2022 – Present

One Medical is a technology-powered primary care organization offering same-day appointments, 24/7 virtual care, and seamless care coordination — aiming to make healthcare more accessible, affordable, and patient-centered.

- Transformed patient scheduling and care plan coordination, creating FHIR-compliant services that support One Medical's same-day appointments, virtual visit follow-ups, and seamless care transitions. Partnered with clinicians and operations teams to capture real-world workflows, ensuring patients could book, reschedule, and receive care updates with minimal friction.

- Built event-driven microservices in .NET 8 and Azure Service Bus to process appointment creation, provider availability updates, and post-visit care notifications in near-real-time — directly reducing missed appointments and improving continuity of care. Implemented idempotency and replay strategies to guarantee patient safety and data consistency.
- Enhanced patient and provider experience by developing Angular 17 components with intuitive UX for scheduling views, provider selection, and telehealth link delivery. Facilitated usability testing with care coordinators and refined designs to minimize cognitive load, which improved booking completion rates.
- Enabled proactive outreach and population health initiatives by embedding MLflow-based risk stratification models into dashboards used by care teams. Configured inference caching and drift alerts to ensure timely, actionable insights for at-risk patients, aligning with One Medical's goal of preventive, not just reactive, care.
- Led infrastructure and platform health initiatives, prioritizing technical debt reduction, refactoring of high-churn services, and performance tuning of patient-facing APIs. Collaborated with product leadership to communicate trade-offs between speed and maintainability, helping to align technical priorities with patient satisfaction metrics.
- Improved system reliability and on-call readiness by implementing OpenTelemetry tracing and Grafana dashboards tied to patient journey KPIs (time-to-book, appointment confirmation latency). Authored runbooks and led incident response simulations, reducing mean time to resolution (MTTR) for patient-impacting incidents.
- Promoted safe and efficient development practices by training engineers on GitHub Copilot and GPT-powered code review assistants, resulting in faster delivery of high-quality features for patient-facing products.
- Acted as a bridge between engineering and clinical teams, translating complex technical concepts into clear, actionable updates during sprint demos and roadmap sessions, enabling business leaders to prioritize initiatives that maximized patient impact and operational efficiency.

Optum (UnitedHealth Group)

Lead Software Engineer | 09/2018 – 07/2022

Optum's Care Delivery & Analytics division builds technology for claims adjudication modernization, provider quality analytics, and population health initiatives supporting millions of patients and providers.

- Re-architected the claims routing and adjudication system, breaking down monolithic code into containerized microservices using .NET Core and gRPC. Conducted deep analysis of legacy workflows with business analysts and compliance officers, creating documentation that mapped every rule and exception. Partnered with QA to develop contract tests ensuring parity between old and new systems.
- Designed Kafka-based streaming pipelines for claims events, introducing schema registry enforcement and backward-compatible schema evolution. Worked with data engineers to design partitioning strategies that balanced throughput and ordering guarantees. Organized knowledge-sharing sessions on distributed stream processing patterns for the wider engineering team.
- Collaborated with UX and provider engagement teams to build React dashboards exposing HCC risk scores, care gap closure rates, and attribution lists. Advocated for accessibility compliance (WCAG) and worked with design teams to add keyboard navigation and screen reader support, ensuring inclusivity for all providers.
- Integrated predictive models by working with MLOps teams to deploy MLflow models with canary releases and drift detection. Facilitated meetings with clinical stakeholders to explain model predictions, gather feedback on false positives, and iterate on threshold tuning.

- Strengthened security posture by implementing OAuth2, JWT token validation, and mTLS between services. Worked hand-in-hand with InfoSec during penetration tests and risk assessments, addressing findings quickly and documenting mitigation steps for audit trails.
- Led infrastructure modernization efforts, writing Terraform modules for AKS clusters, implementing CI/CD pipelines in Azure DevOps, and collaborating with SRE to set up auto-scaling, liveness probes, and health checks. This effort reduced manual maintenance and increased deployment reliability.
- Mentored junior engineers, performing detailed code reviews, pair programming on critical services, and running weekly architecture study groups. Introduced SonarQube quality gates, AI-assisted static analysis, and automated dependency scanning, increasing codebase health and reducing production issues.
- Actively participated in quarterly product roadmap planning, working with leadership to balance compliance-driven initiatives with revenue-generating features. Presented architecture proposals to executive committees, building consensus across product, compliance, and engineering groups.

WellSky

Software Engineer | 01/2015 – 08/2018

WellSky provides post-acute care and home health solutions, supporting hospitals, rehab centers, and home health agencies in scheduling, documentation, and compliance.

- Modernized the scheduling module, refactoring ASP.NET MVC pages into a single-page Angular application with improved performance and usability. Partnered with clinicians and scheduling coordinators to uncover workflow pain points, such as overlapping visits and multi-provider resource conflicts, and built validation logic directly into the UI.
- Developed GraphQL and REST APIs to expose patient encounter data to external pharmacy and lab systems. Worked closely with interoperability teams to reconcile HL7 variations, adding automated validation tests for message conformance and error recovery routines.
- Built Python ETL pipelines in Airflow to normalize HL7 ADT messages, handle duplicate detection, and enrich records with master patient index data. Documented data lineage and collaborated with data governance teams to ensure accuracy.
- Added real-time WebSocket notifications with Redis-backed queues, solving ordering issues with FIFO guarantees. Collaborated with network teams to ensure websocket persistence through load balancers and tested failover scenarios.
- Enhanced security and compliance by enabling encryption-at-rest with SQL Server TDE and implementing TLS 1.2. Integrated Trivy vulnerability scans and dependency checks into CI/CD pipelines.
- Expanded automated test coverage using xUnit and Postman, integrating them with nightly regression runs. Created smoke tests and mock data sets to catch data contract violations earlier in the SDLC.
- Actively engaged in sprint ceremonies and stakeholder demos, translating clinician requests into technical stories, negotiating scope with product managers, and advocating for usability improvements that reduced cognitive burden on nurses.

Education

Bachelor of Science in Computer Science | Tufts University | 2010 - 2014